

KOSZULNESS CHART FUNCTORS AND CONDITIONAL ATLAS CRITERIA: THE KONTSEVICH–TAMARKIN REFORMULATION OF CHIRAL KOSZUL DUALITY

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ABSTRACT. For a finite-PBW chiral algebra $\mathcal{A}$ on an algebraic curve X , the Kontsevich–Tamarkin action organizes Koszulness tests into associator-dependent chart functors. After choosing a framing and a finite PBW-Artin stage, the Koszul locus is represented by an open subfunctor of the finite-stage parameter space; on nonempty framed components the chart transitions form the usual pro-GRT₁ transfer torsor. This paper records the chart formulation and keeps the logical directions separate. The algebraic/PBW core gives the bidirectional Koszulness criteria. Hochschild, factorization-homology, Lagrangian, and mixed-Hodge criteria are consequences or conditional converses under the hypotheses stated in the theorem. The scalar degree-two saturation statement is proved only on the scalar named rows; outside those rows it remains a conditional chart assertion.

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1. INTRODUCTION: THE KOSZUL LOCUS AND WHY IT MATTERS

1.1. **The problem.** The bar complex of a chiral algebra $\mathcal{A}$ on an algebraic curve X is the cofree E_1 -coalgebra

$$\bar{B}^{\text{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$$

with deconcatenation coproduct, where $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal and $|s^{-1}v| = |v| - 1$ is the desuspension grading. The bar differential extracts OPE data through the logarithmic propagator $\eta = d \log(z_1 - z_2)$ on Fulton–MacPherson configuration spaces $\overline{C}_n(X)$. This construction exists for every chiral algebra. Its differential squares to zero. Its cohomology defines a factorization coalgebra. The cobar functor Ω^{ch} produces from $\bar{B}^{\text{ch}}(\mathcal{A})$ a new chiral algebra, and the counit $\varepsilon_{\mathcal{A}}: \Omega^{\text{ch}}(\bar{B}^{\text{ch}}(\mathcal{A})) \rightarrow \mathcal{A}$ is a natural transformation.

The question is: when is $\varepsilon_{\mathcal{A}}$ a quasi-isomorphism?

When it is, the bar complex is a complete invariant: it encodes $\mathcal{A}$ up to quasi-isomorphism, the spectral sequences computing its cohomology collapse, the resolution is linear in the PBW degree, and a single scalar $\kappa(\mathcal{A})$ controls the universal Maurer–Cartan element at degree two. When it is not, the bar–cobar object persists only in the coderived category, the spectral sequences have nontrivial higher differentials, and the resolution is nonlinear in a way that resists finite description.

The locus on which $\varepsilon_{\mathcal{A}}$ is a quasi-isomorphism is the *Koszul locus*. This paper gives fourteen tests for this locus. The algebraic/PBW tests are bidirectional criteria. The Hochschild, factorization-homology, Lagrangian, and mixed-Hodge tests enter with the hypotheses and logical directions stated below.

1.2. Context. In classical algebra, the Koszul property of a quadratic algebra $\mathcal{A} = T(V)/(R)$ has been characterized by Priddy [25] (purity of the bar complex), Beilinson–Ginzburg–Soergel [13] (Ext diagonal vanishing), and Loday–Vallette [22] ($\mathcal{A}_{\infty}$ -formality of the Koszul dual cooperad). Each formulation detects the same phenomenon from a different mathematical vantage point: the bar complex has diagonal bar cohomology exactly when it is concentrated in degree one.

Chiral algebras are not quadratic. The Virasoro algebra has a quartic OPE pole; $\mathcal{W}$ -algebras have poles of arbitrarily high order; Yangians have spectral-parameter relations. None of these admit a quadratic presentation, and the Beilinson–Ginzburg–Soergel theory does not see them directly. Three features of chiral algebras on algebraic curves defeat the classical framework simultaneously: (a) the bar differential on the genus expansion satisfies $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, so the classical bar–cobar adjunction no longer applies beyond genus zero; (b) the tensor power $V^{\otimes n}$ is replaced by the factorization product $\mathcal{A}^{\boxtimes n}$ over $\overline{C}_n(X)$, and the differential is a sum of OPE residues at boundary strata, not a combinatorial sum of face maps; (c) conformal-weight components are infinite-dimensional, so convergence of spectral sequences is a separate problem.

The structure of chiral objects forces the extension of Koszul theory beyond the classical quadratic setting. The Francis–Gaitsgory adjunction $\text{Lie}^{\text{ch}} \dashv \text{Com}^{\text{ch}}$ on $\text{Ran}(X)$ [15] settles Koszul duality of the Lie–Com pair at the operadic level, but it does not determine whether a specific chiral algebra is Koszul: it does not see the genus corrections in the bar complex, and it does not detect the shadow obstruction tower whose depth varies from two to infinity across the standard families.

1.3. The fourteen characterizations. The main theorem of this paper assembles fourteen tests for chiral Koszulness. They sort into four groups by mathematical origin.

Homological algebra (from the bar complex and its spectral sequences): chiral Koszulness (i), PBW E_2 -collapse (ii), $\mathcal{A}_{\infty}$ -formality (iii), Ext diagonal vanishing (xii), Kac–Shapovalov determinant nonvanishing (xiii).

Deformation theory (from the convolution algebra and its formality): shadow-tower formality (iv), E_2 -formality of chiral Hochschild cohomology (v), Sklyanin $H^2 = 0$ (xiv).

Factorization geometry (from configuration spaces and the Ran space): curve independence (vi), PBW universality (vii), Barr–Beck–Lurie monadicity (viii), factorization homology concentration (ix), FM boundary acyclicity (x).

Tropical geometry (from the dual complex of the FM compactification): tropical Koszulness (xi).

Two further conditions stand outside these fourteen. The *Lagrangian criterion* (xv): $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1) -shifted symplectic deformation space, proved equivalent under a perfectness hypothesis. And *$\mathcal{D}$ -module purity* (xvi): each $\tilde{B}_n^{\text{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module, proved to imply Koszulness with the converse established for the affine Kac–Moody family and open in general.

1.4. The six-fold TFAE on the Koszul locus, with a class- G seventh equivalence: hub-and-spoke reconstitution. The fourteen characterizations above form an atlas of Koszulness chart functors. At genus zero, a strictly smaller sub-atlas admits a sharper presentation: six characterizations are mutually equivalent through a single hub on the full Koszul locus, and a seventh (SC-formality) is equivalent to the same hub when attention is restricted to the class- G stratum.

Theorem 1.1 (Six-fold TFAE at genus zero, with a class- G seventh equivalence; hub-and-spoke). *Let $\mathcal{A}$ be an augmented chiral algebra on a smooth genus-zero curve X equipped with a PBW filtration $F_\bullet$, and let $\tau_{\text{univ}}: \bar{B}^{\text{ch}}(\mathcal{A}) \rightarrow \mathcal{A}$ be the universal twisting morphism of the canonical bar twisting morphism $\tau_{\text{univ}}: \bar{B}^{\text{ch}}(\mathcal{A}) \rightarrow \mathcal{A}$. Conditions (S1)–(S6) below are equivalent through the PBW E_2 -collapse hub on the full Koszul locus; condition (S7) is equivalent to (S6) on the class- G stratum and otherwise fails.*

- (S1) Chiral Koszul morphism: τ_{univ} is a chiral Koszul morphism (Definition 3.1).
- (S2) Counit quasi-isomorphism: the counit $\varepsilon_\tau: \Omega_X(\bar{B}^{\text{ch}}(\mathcal{A})) \rightarrow \mathcal{A}$ is a quasi-isomorphism.
- (S3) Unit weak equivalence: the unit $\eta_\tau: \bar{B}^{\text{ch}}(\mathcal{A}) \rightarrow \bar{B}^{\text{ch}}(\Omega_X(\bar{B}^{\text{ch}}(\mathcal{A})))$ is a weak equivalence of conilpotent complete factorisation coalgebras.
- (S4) Twisted tensor acyclicity: both twisted tensor products $K_\tau^L(\mathcal{A}, \bar{B}^{\text{ch}}(\mathcal{A}))$ and $K_\tau^R(\bar{B}^{\text{ch}}(\mathcal{A}), \mathcal{A})$ are acyclic.
- (S5) Bar concentration in weight 1: bar cohomology $H^{p,q}(\bar{B}^{\text{ch}}(\mathcal{A}))$ vanishes for $q \neq 0$, with $H^{p,0}$ concentrated in bar-weight p .
- (S6) PBW E_2 -collapse (the hub): the PBW spectral sequence of $\bar{B}^{\text{ch}}(\mathcal{A})$ collapses at E_2 : $d_r = 0$ for all $r \geq 2$.
- (S7) SC-formality at genus zero (class- G scoped): the transferred Swiss-cheese operation tower has $m_k^{\text{SC}} = 0$ for all $k \geq 3$.

(Hub-and-spoke structure.) *The hub is (S6). For each $i \in \{1, 2, 3, 4, 5\}$ there is a single proved bidirection $i \Leftrightarrow (\text{S6})$; the six-fold TFAE on the Koszul locus follows by transitivity from these five spokes. A sixth, class- G -scoped spoke (S7) $\Leftrightarrow (\text{S6})$ furnishes the seventh equivalence on that stratum.*

(Load-bearing bidirection.) *The spoke (S4) $\Leftrightarrow (\text{S6})$ (twisted-tensor $\Leftrightarrow$ PBW) is the substantive bidirection: both forward and reverse require genuine theorems (PBW implies twisted-tensor acyclicity through the free resolution; twisted-tensor acyclicity implies PBW through degeneration of the filtration spectral sequence). The affine Kac–Moody family $V_k(\mathfrak{sl}_2)$ at generic k witnesses both implications concretely.*

(Class- G restriction of (S7).) *Spoke (S7) is class- G scoped: it is equivalent to (S6) only when $\mathcal{A}$ is class G . Class L, C, M algebras are chirally Koszul (all six non-(S7) spokes hold) yet fail SC-formality (compare Remark 1.4).*

(Higher-genus extension.) *Spokes (S1)–(S4) survive verbatim to $g \geq 1$ within the Koszul locus; spoke (S5) survives to $g \geq 1$ only for class G , since $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ obstructs weight-1 concentration off class G .*

Proof sketch. The six spokes are proved elsewhere: spoke (S1) by Definition 3.1 together with the Koszul characterisation through PBW E_2 -collapse; spoke (S2) is the bar–cobar counit criterion, which is equivalent to E_2 -collapse through the twisting morphism dictionary; spoke (S3) by the conilpotent-complete universal property of $\bar{B}^{\text{ch}}$; spoke (S4) by the substantive PBW bidirection cited above; spoke (S5) by Priddy’s criterion in its chiral avatar; spoke (S7) by the class- G SC-formality criterion in the parent monograph. The transitivity closure is the elementary observation that six proved bidirections through a common hub determine the remaining 15 bidirections of the 21-arrow complete graph. $\square$

Remark 1.2 (Why the hub is PBW E_2 -collapse). The PBW spectral sequence is the universal invariant that compares the associated graded algebra of $\mathcal{A}$ (determined by the filtration alone, independent of the OPE) with the bar cohomology of $\mathcal{A}$ (determined by the OPE, with the filtration tracking PBW depth). Its E_2 -collapse is the locus where these two pieces of data agree. Every classical criterion reduces to this comparison; all sit at a distance of one spoke from the hub.

1.5. Two distinctions.

Remark 1.3 (Shadow depth and Koszulness). The four complexity classes $G/L/C/M$, with shadow depths $\{2, 3, 4, \infty\}$, record the degree at which the shadow obstruction tower first becomes nontrivial.

They do *not* record failure of Koszulness. Every standard chiral algebra is chirally Koszul. The Virasoro algebra has shadow depth $r_{\max} = \infty$ (class M); it is chirally Koszul at generic central charge. Shadow depth classifies complexity *within* the Koszul world.

Remark 1.4 (Koszulness and SC-formality). Koszulness is the condition that bar cohomology is concentrated in degree one; SC-formality is the condition that the higher operations $m_k^{\text{SC}} = 0$ for $k \geq 3$ in the Swiss-cheese bar complex. All standard families are chirally Koszul; only class G (Heisenberg) is SC-formal. Confusing the two misidentifies what the shadow tower measures.

1.6. Conventions. All chain complexes are cohomologically graded: $|d| = +1$. Desuspension lowers degree: $|s^{-1}v| = |v| - 1$. The bar complex uses the augmentation ideal $\bar{\mathcal{A}} = \ker(\varepsilon)$; we write $\bar{B}^{\text{ch}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ with deconcatenation coproduct. We work in characteristic zero over an algebraically closed field. The Fulton–MacPherson compactification $\bar{C}_n(X)$ is the real oriented blowup along all diagonals; it is a manifold with corners whose boundary strata are indexed by rooted trees [16]. We cite [23] for background results established in the parent monograph.

2. KOSZULNESS CHART FUNCTORS

2.1. Principle. The classical literature presents Koszulness through lists of equivalent conditions. That presentation identifies a property with the charts that detect it. In the chiral setting each chart is tied to a choice of Drinfeld associator. Tamarkin transfer carries a chart at one associator to a chart at another. The natural object is therefore a chart functor: finite-stage algebraic on PBW-Artin truncations, and pro-algebraic only after passing to the full associator torsor.

Definition 2.1 (Koszulness chart functor). Let $\mathcal{A}$ be a chiral algebra on X with PBW filtration, and fix a finite PBW-Artin stage N together with a framing of its associated graded. The *Koszulness chart functor* $\mathcal{M}_{\text{Kosz}}^{\leq N, \text{fr}}(\mathcal{A})$ is the functor

$$\mathcal{M}_{\text{Kosz}}^{\leq N, \text{fr}}(\mathcal{A}) : \mathbf{CAlg}_{\mathbb{Q}} \longrightarrow \mathbf{Set}, \quad R \longmapsto \{(\Phi, c_{\Phi}) : \Phi \in \text{Assoc}(R), c_{\Phi} \in \text{Chart}_{\Phi}(\mathcal{A}; R)\}$$

where $\text{Assoc}(R)$ is the torsor of Drinfeld associators [4] over R , and $\text{Chart}_{\Phi}(\mathcal{A}; R)$ is the set of *chart coordinates*: triples $(\Pi_{\Phi}, \text{wit}_{\Phi}, \gamma_{\Phi})$ consisting of

- (a) a Koszulness predicate Π_{Φ} depending only on the pair $(\mathcal{A}, \Phi)$;
- (b) a witness $\text{wit}_{\Phi}(\mathcal{A})$ in an appropriate test category, providing a proof witness for $\Pi_{\Phi}(\mathcal{A})$;
- (c) a transition 1-cocycle $\gamma_{\Phi} : \Phi \rightsquigarrow \Phi'$ to every other associator, satisfying the Tamarkin compatibility [12]

$$\Pi_{\Phi}(\mathcal{A}) \iff \Pi_{\Phi \cdot g}(\mathcal{A}) \quad \forall g \in \text{GRT}_1(R).$$

Theorem 2.2 (Finite-stage Koszulness atlas). (Kontsevich–Tamarkin reformulation.) *Let $\mathcal{A}$ be a positive-energy chiral algebra on X with finite PBW windows. After choosing a finite PBW-Artin stage and a framing, the Koszul locus is represented by an open subfunctor of the finite-stage parameter space. On every nonempty framed component, the transition functors between associator charts form the usual pro-GRT₁ transfer torsor. More precisely:*

- (A1) *The finite-stage chart functor is nonempty precisely when the chosen stage contains a witness for the chiral Koszul morphism of Definition 3.1. Passing to the inverse limit over PBW-Artin stages recovers chiral Koszulness under the complete separated Mittag–Leffler hypothesis.*
- (A2) *The forgetful map $\pi : \mathcal{M}_{\text{Kosz}}^{\leq N, \text{fr}}(\mathcal{A}) \rightarrow \text{Assoc}$ carries a pro-GRT₁-transfer action on each nonempty framed component.*

- (A3) The fourteen classical characterizations (i)–(xiv) give a finite chart atlas $\{U_j\}_{j=1}^{14}$: charts $U_1, \dots, U_{10}$ are centred at the KZ associator Φ_{KZ} ; U_{11} is centred at Φ_{AT} ; U_{12} at $\Phi_{\text{dR}, \text{B}}$; U_{13} at Φ_{ell} ; U_{14} at Φ_{Kon} . Intersections are identified by Tamarkin transfer whenever both charts contain a Koszul witness.
- (A4) The shadow-class stratification $G \sqcup L \sqcup C \sqcup M \sqcup FF$ (Feigin–Frenkel critical stratum) selects a distinguished chart on each stratum: Φ_{KZ} on G and on the generic Kac–Moody locus within L , Φ_{ell} on L , Φ_{AT} on C , $\Phi_{\text{Kon}} = \Phi_{\infty}$ on M , and $\Phi_{\text{dR}, \text{B}}$ on FF .

Proof. (A1). Definition 3.1 is GRT_1 -invariant: acyclicity of the twisted tensor product $K_{\tau}^L(\mathcal{A}, \bar{B}^{\text{ch}}(\mathcal{A}))$ is preserved under the Tamarkin transfer $K_{\tau}^L \mapsto K_{\Phi, \tau}^L$, because the transfer is a quasi-isomorphism (Kontsevich [10], Tamarkin [12]). At a fixed finite PBW–Artin stage this gives an open condition on the framed parameter space: the determinant minors for the two twisted tensor complexes must be invertible. Taking the inverse limit over finite windows gives the full Koszul predicate when the PBW filtration is complete, separated, and satisfies the Mittag–Leffler condition.

(A2). Fix $\Phi_0 = \Phi_{\text{KZ}}$ as base point. For any other Φ , the Grothendieck–Teichmüller torsor provides a unique $g_{\Phi} \in \text{GRT}_1$ with $\Phi = \Phi_0 \cdot g_{\Phi}$. The Tamarkin transfer $g_{\Phi}^*: \text{Chart}_{\Phi_0}(\mathcal{A}) \rightarrow \text{Chart}_{\Phi}(\mathcal{A})$ is an equivalence on each framed component containing a Koszul witness, and the cocycle condition $g_{\Phi''} = g_{\Phi'} \cdot h_{\Phi' \rightarrow \Phi''}$ holds because associator composition is strictly associative in GRT_1 . Thus the chart transitions carry the expected pro- GRT_1 torsor structure. The global object is a pro-algebraic chart functor assembled from finite PBW stages.

(A3). The ten classical characterizations of Theorem 3.2 (i)–(x) sit at Φ_{KZ} : each is stated in the KZ language (PBW spectral sequence, bar cohomology, $\mathcal{A}_{\infty}$ -formality, BBL monadicity, etc.) whose coherence data is the KZ associator. The four supplementary charts $U_{11}, \dots, U_{14}$ are constructed in Propositions 2.4, 2.5, 2.6, and 2.7 below, with the hypotheses stated in those propositions. Non-emptiness of a pairwise intersection means that both charts contain a Koszul witness for the same finite PBW stage. If $\Phi_2 = \Phi_1 \cdot g$, Tamarkin transfer by $g \in \text{GRT}_1$ identifies the two chart functors on that component.

(A4). The shadow class is an intrinsic invariant of the MC element $\Theta_{\mathcal{A}}$; the selection of a distinguished chart on each stratum is determined by the computational complexity of the shadow algebra bracket $[-, -]^{\text{sh}}$ there. Class G: abelian shadow algebra, KZ associator is the canonical flat connection on $\overline{\mathcal{C}}_n(\mathbb{C})$, and U_1 (PBW E_2 -collapse) is trivially satisfied. Class L: cubic bracket only, genus-one Enriques coupling; the elliptic associator Φ_{ell} with its modular equivariance is the natural chart. Class C: contact termination at S_4 , AT Lagrangian coordinates diagonalize the quartic bracket (Proposition 2.4). Class M: infinite tower, Kontsevich integral gives chart coordinates tracking the full MC expansion (Proposition 2.7). Class FF: Feigin–Frenkel locus $k = -h^{\vee}$; the de Rham–Betti chart $\Phi_{\text{dR}, \text{B}}$ sees the purity via BGG resolution (Proposition 2.5). $\square$

Remark 2.3 (Kontsevich–Tamarkin reformulated). The classical Kontsevich–Tamarkin theorem [10, 12] is usually stated: “formality of little disks holds universally, but the choice of formality isomorphism depends on the associator.” The Koszulness chart functor gives this statement operational content at the level of a single algebra: chiral Koszulness is a GRT_1 -invariant property of $\mathcal{A}$; the classical characterizations are GRT_1 -coordinate charts on the finite PBW–Artin chart functors. Switching between two classical criteria is an application of the Tamarkin Φ -transfer; the transfer becomes trivial on the GRT_1 -invariants.

2.2. Atlas. The atlas has fourteen charts. The first ten are the classical conditions (the old Theorem 2.2, stated below with its original numbering as Theorem 3.2); the additional four are finite-PBW comparison charts. They are independent chart coordinates, but their converses require the perfectness, purity, genus-one flatness, or two-colour operadic hypotheses stated below.

Proposition 2.4 (U_{11} : Alekseev–Torossian chart). *At the Alekseev–Torossian associator $\Phi_{\text{AT}} [1]$, chiral Koszulness of a chiral algebra $\mathcal{A}$ with PBW filtration is equivalent to Lagrangian transversality of $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^\dagger}$ in the (-1) -shifted symplectic AT deformation space $\mathcal{M}_{\text{comp}}^{\text{AT}}$, provided the ambient tangent complex is two-term perfect and the AT transfer preserves the finite PBW filtration.*

Proof sketch. The Alekseev–Torossian associator is constructed from the Kashiwara–Vergne equations on free Lie algebras [1]: it provides an explicit homotopy between the KZ associator and its symmetrization, together with a canonical choice of Baker–Campbell–Hausdorff basis on the target. Transfer the (-1) -shifted symplectic deformation space $\mathcal{M}_{\text{comp}}$ of the classical Lagrangian criterion to $\mathcal{M}_{\text{comp}}^{\text{AT}}$ using the AT gauge:

$$\Psi_{\text{AT}}: \mathcal{M}_{\text{comp}} \xrightarrow{\sim} \mathcal{M}_{\text{comp}}^{\text{AT}}, \quad \Psi_{\text{AT}}^* \omega_{\text{AT}} = \omega,$$

where $\omega, \omega_{\text{AT}}$ are the respective symplectic forms. The AT gauge is constructed so that the ambient tangent complex of $\mathcal{M}_{\text{comp}}^{\text{AT}}$ is two-term perfect: concentrated in degrees $\{1, 2\}$, with generators the Kashiwara–Vergne F, H pair and relations the Duflo–Kontsevich isomorphism. The chart is therefore available on the finite-PBW component where this two-term perfect model exists. Lagrangian transversality of $\Psi_{\text{AT}}(\mathcal{M}_{\mathcal{A}})$ and $\Psi_{\text{AT}}(\mathcal{M}_{\mathcal{A}^\dagger})$ is equivalent to acyclicity of the AT-twisted tensor product $K_{\tau}^{\text{AT}}(\mathcal{A}, \mathcal{A}^\dagger)$, which equals the KZ-twisted product up to a quasi-isomorphism induced by the AT gauge; hence Koszulness. $\square$

Proposition 2.5 (U_{12} : Hodge–Betti chart). *At the de Rham–Betti associator $\Phi_{\text{dR,B}} [3]$, chiral Koszulness of a chiral algebra $\mathcal{A}$ with conformal weight grading is equivalent to $\mathcal{D}$ -module purity of each $\bar{B}_n^{\text{ch}}(\mathcal{A})$ as a mixed Hodge module (condition (xvi)) in the families where the BGG or DS comparison supplies a pure finite PBW resolution. In general, purity implies FM boundary acyclicity; the converse remains a family-specific assertion.*

Proof sketch. The de Rham–Betti associator sits at the boundary point of the Drinfeld associator torsor where the KZ connection on $\overline{\mathcal{C}}_n(\mathbb{C})$ is identified with the comparison isomorphism between de Rham and Betti cohomology of a mixed Hodge structure [3]. At this base point, chart coordinates are weight-filtered mixed Hodge structures on the bar complex.

For any $\mathcal{A}$ with polynomial PBW associated graded, the Hodge–Betti chart detects purity directly: the weight filtration on $\bar{B}_n^{\text{ch}}(\mathcal{A})$ is recovered from the Hodge–Betti period matrix, and purity means the weight-filtered piece agrees with the pure part. The implication (xvi) $\Rightarrow$ (x) is classical (Saito’s theorem on mixed Hodge modules). The converse is proved only where an independent pure resolution is available: for $\mathcal{A} = \text{Vir}_c$ at generic c , the BGG resolution of the vacuum Verma module [6, 11] provides a pure resolution of the bar complex; combined with non-degeneracy $\det G_h \neq 0$ outside the discrete minimal-model lattice, this gives purity directly. The same argument applies to the simple $\mathcal{W}$ -rows for which the DS reduction preserves the relevant pure PBW resolution. $\square$

Proposition 2.6 (U_{13} : Enriquez elliptic chart). *At the Enriquez elliptic associator $\Phi_{\text{ell}} [5]$, chiral Koszulness of $\mathcal{A}$ is equivalent to flatness of the genus-one twisted Künneth formula $\bar{B}^{\text{ch}}(\mathcal{A} \boxtimes_{\text{ell}} \mathcal{A}^\dagger) \simeq \bar{B}^{\text{ch}}(\mathcal{A}) \boxtimes \bar{B}^{\text{ch}}(\mathcal{A}^\dagger)$ on the elliptic configuration space $\overline{\mathcal{M}}_{1,n}$. This is a genus-one chart criterion on the finite PBW component where the KZB transfer and the elliptic PBW spectral sequence are defined.*

Proof sketch. The Enriquez elliptic associator [5] is constructed from the KZB connection on the universal elliptic curve; it provides a canonical elliptic completion of the KZ associator, with modular group $\text{SL}_2(\mathbb{Z})$ -equivariance built in. The chart coordinate at Φ_{ell} is the elliptic twisted Künneth isomorphism: for chirally Koszul $\mathcal{A}$, the genus-one bar complex factorises as a Künneth product of genus-zero data twisted by the elliptic R-matrix.

Flatness of the twisted Künneth is equivalent to E_2 -collapse of the genus-one PBW spectral sequence on the elliptic curve. This is condition (ii) transported to $\overline{C}_n(E)$; the classical proof on $\overline{C}_n(\mathbb{C})$ carries over, using the fact that the elliptic R-matrix (Felder 1994) is the image of the rational R-matrix under the Tamarkin transfer from Φ_{KZ} to Φ_{ell} . Hence U_{13} is witnessed on the elliptic finite-PBW component; the chart distinguishes the class L stratum because affine Kac–Moody algebras at generic level are where the genus-one Enriquez coupling first activates (cubic bracket S_3 on the shadow algebra). $\square$

Proposition 2.7 (U_{14} : Kontsevich integral chart). *At the Kontsevich integral associator Φ_{Kon} [9], chiral Koszulness of $\mathcal{A}$ is equivalent to homotopy-Koszulity of the Swiss-cheese operad $SC^{ch,top}$ acting on the pair $(\mathcal{Z}_{ch}^{der}(\mathcal{A}), \mathcal{A})$ at every degree of the MC expansion, assuming a two-colour chain-level Kontsevich–Tamarkin formality comparison for the chosen finite-PBW component. On the class M stratum, this chart records the infinite shadow tower in chord-diagram coordinates.*

Proof sketch. The Kontsevich integral [9] is the universal Vassiliev invariant: it assembles into an associator Φ_{Kon} at the Vassiliev base point, with chart coordinates indexed by chord diagrams. For chiral Koszul $\mathcal{A}$, the MC element $\Theta_{\mathcal{A}}$ is a sum over chord diagrams in the Kontsevich expansion, each coefficient computed by the shadow algebra bracket.

Homotopy-Koszulity of $SC^{ch,top}$ on the pair $(\mathcal{Z}_{ch}^{der}(\mathcal{A}), \mathcal{A})$ is the condition that the bar complex of $SC^{ch,top}$ has cohomology concentrated on the operadic diagonal. At the Kontsevich base point, this is the assertion that chord diagram coefficients satisfy the IHX and STU relations, which is automatic for $SC^{ch,top}$ by Kontsevich formality for the little disks in two colors, once the two-colour chain-level comparison is fixed. On class M, the Virasoro shadow tower is an infinite sum over chord diagrams indexed by the tower levels, and $SC^{ch,top}$ homotopy-Koszulity is precisely the statement that this sum converges in the coderived category [24]. $\square$

Remark 2.8 (Chart redundancy and the torsor structure). The fourteen charts are not equal in logical strength. The core PBW charts give bidirectional criteria; the supplementary charts give consequences or conditional converses. The content of Theorem 2.2 (A2) is that the overlap between valid charts is organized by the GRT_1 -torsor structure, and the content of (A4) is that the redundancy is broken on each stratum by the distinguished chart. Each chart is adapted to a stratum, and finiteness here means finite after choosing the named comparison base points, not finite presentation of the full associator torsor.

2.3. Virasoro non-circular closure. The classical proof of Virasoro Koszulness at generic c traced a bar–cobar circle: Koszulness implies E_2 -collapse implies concentration of chiral Hochschild cohomology, and the converse closes the loop through bar–cobar inversion. Every step uses the bar complex; circularity is latent. The Hodge–Betti chart U_{12} provides an independent proof via Feigin–Fuchs BGG resolution.

Theorem 2.9 (Virasoro Koszulness, non-circular). *For the Virasoro vertex algebra Vir_c at generic central charge $c \notin \{c_{p,q} : (p, q) \in \mathbb{Z}_{>0}^2, \gcd(p, q) = 1\}$, the bar cohomology weight components satisfy*

$$H^\bullet(Vir_c, Vir_c)_{\text{weight } r+1} = 0 \quad \text{for all } r \geq 2,$$

using the Hodge–Betti comparison. Consequently, Vir_c is chirally Koszul, proved directly at the U_{12} chart.

Proof. We work entirely at the Hodge–Betti chart U_{12} . The proof has two classical inputs.

Step 1: Feigin–Fuchs BGG resolution. For each highest weight h in the vacuum Verma module $M_o(c)$ of Vir_c , the Feigin–Fuchs theorem [6] (see also Rocha-Caridi–Wallach [11]) constructs a resolution of the vacuum irreducible by Verma modules indexed by elements of an affine Weyl group acting on weights:

$$\cdots \rightarrow \bigoplus_{w \in W^{(2)}} M_{w \cdot o}(c) \rightarrow \bigoplus_{w \in W^{(1)}} M_{w \cdot o}(c) \rightarrow M_o(c) \rightarrow L(c, o) \rightarrow 0,$$

with differentials given by singular vector embeddings. At generic c (outside the minimal-model lattice), each singular vector embedding is nonzero and the resolution is exact. Each term is a Verma module, whose mixed Hodge structure is pure of weight given by its conformal weight (this is classical; Verma modules are free over the PBW algebra).

Step 2: Kac determinant non-degeneracy. The Kac determinant [6, 8] at conformal weight h is

$$\det G_h = \prod_{\substack{p,q \geq 1 \\ pq \leq h}} (h - h_{p,q}(c))^{P(h-pq)},$$

where P is the partition function and $h_{p,q}(c) = ((p\alpha_+ + q\alpha_-)^2 - (\alpha_+ + \alpha_-)^2)/4$ with $\alpha_{\pm}$ the Liouville parameters determined by c . For c outside the minimal-model lattice, the Kac determinant is nonzero at every bar-relevant weight h : the BGG resolution of Step 1 embeds faithfully in the bar complex.

Step 3: Purity of the bar complex. Combine Steps 1 and 2. The BGG resolution of Step 1 gives a $\mathcal{D}$ -module resolution of the unit ω_X of Vir_c by a sum of Verma modules; each Verma module is pure of its conformal weight in the Hodge–Betti category (Verma modules have free PBW associated graded, hence polynomial mixed Hodge structure). Step 2 (Kac determinant non-degeneracy at generic c) guarantees that the BGG differentials are injective at every weight, so the resolution is exact. The bar complex $\bar{B}_n^{\text{ch}}(\text{Vir}_c)$ is quasi-isomorphic to the n -fold iterate of this resolution, and iterated pure objects remain pure by Saito’s stability theorem [26].

Step 4: Weight vanishing. Purity of $\bar{B}_n^{\text{ch}}(\text{Vir}_c)$ of weight n forces $H^\bullet(\text{Vir}_c, \text{Vir}_c)_{\text{weight } r+1} = 0$ for every $r \geq 2$ off the Koszul diagonal: any nonzero class in weight $r+1 \neq n$ would contradict purity of the bar complex in weight n after the Ext computation.

Conclusion. The four steps use only Feigin–Fuchs BGG (Step 1), Kac determinant (Step 2), Saito stability (Step 3), and purity vanishing (Step 4). $\square$

Remark 2.10 (Non-circular input). The Hodge–Betti proof uses data external to bar–cobar inversion: the Feigin–Fuchs BGG resolution and the Kac determinant. The bar complex receives that input through purity of the induced $\mathcal{D}$ -module resolution.

2.4. Yangian chart inclusion. The associative Koszulness of the Yangian $Y(\mathfrak{g})$ uses the classical Polishchuk–Positselski PBW criterion. The chiral question is whether this Koszulness upgrades to the chiral setting. The chart-functor formulation isolates the precise comparison map.

Theorem 2.11 (Yangian chart inclusion). *For the Yangian $Y(\mathfrak{g})$ with $\mathfrak{g}$ simple, assume that the RTT evaluation data define a finite-PBW E_1 -chiral avatar and that the chart-comparison functor is representable and reflects the Shapovalov open locus. Then every such associative PBW chart maps to the corresponding chiral Koszulness chart:*

$$\mathcal{M}_{\text{Kosz}}^{\text{assoc}}(Y) \hookrightarrow \mathcal{M}_{\text{Kosz}}^{\text{chiral}}(Y).$$

This map is a GRT_1 -equivariant open immersion on the comparison component. Chart-surjectivity remains open in higher rank.

Proof of the embedding. For $\Phi = \Phi_{\text{KZ}}$, the associative PBW chart U_i detects $Y(\mathfrak{g})$ -Koszulness in the classical PBW sense. Transport this chart coordinate to the chiral setting through the assumed finite-PBW E_1 -chiral avatar: the RTT data defines the required formal-disk chiral structure via the evaluation modules V_u , and the PBW filtration is compatible with the chiral bracket. The associative chart coordinate $(\Phi_{\text{KZ}}, c_{\text{KZ}}^{\text{assoc}})$ maps to the chiral chart coordinate $(\Phi_{\text{KZ}}, c_{\text{KZ}}^{\text{chiral}})$: both detect E_2 -collapse of the same PBW spectral sequence, the associative one on $Y(\mathfrak{g})$ and the chiral one on its E_1 -chiral avatar. Hence $\mathcal{M}_{\text{Kosz}}^{\text{assoc}}(Y) \hookrightarrow \mathcal{M}_{\text{Kosz}}^{\text{chiral}}(Y)$ is defined; GRT_1 -equivariance of both sides makes it an equivariant map.

By hypothesis, the associative chart U_1^{assoc} is the complement of the zero locus of the Shapovalov determinant, and its preimage in $\mathcal{M}_{\text{Kosz}}^{\text{chiral}}(Y)$ is the complement of the same zero locus. Hence the comparison morphism is open on this component. $\square$

Remark 2.12 (Associative and chiral Yangian charts). The chart formulation separates associative Koszulness from chiral Koszulness. The associative atlas embeds in the chiral atlas on the comparison component. For $\mathfrak{g} = \mathfrak{sl}_2$ the embedding is an isomorphism; for higher rank, chart-surjectivity is the remaining open question.

3. THE FOURTEEN CHARACTERIZATIONS

Let $\mathcal{A}$ be a chiral algebra on a smooth projective curve X , equipped with its PBW filtration $F_\bullet$. Write $\bar{B}^{\text{ch}}(\mathcal{A}) = T^c(\mathcal{J}^{-1}\bar{\mathcal{A}})$ for the geometric chiral bar complex with deconcatenation coproduct, $R_{\mathcal{A}} = \text{gr}^F \mathcal{A}$ for the associated graded (a Poisson vertex algebra), and Ω^{ch} for the cobar functor.

Definition 3.1 (Chiral Koszul morphism). A chiral twisting datum $(\mathcal{A}, C, \tau, F_\bullet)$ consisting of an augmented chiral algebra $\mathcal{A}$, a conilpotent factorization coalgebra C , a degree-+1 Maurer–Cartan element $\tau: C \rightarrow \mathcal{A}$ in the convolution algebra, and a compatible PBW filtration, is *Koszul* if the following three conditions hold:

- (a) $\text{gr}_F \mathcal{A}$ is classically Koszul on each finite PBW window;
- (b) both twisted tensor products

$$K_\tau^L(\mathcal{A}, C) = (\mathcal{A} \otimes C, d_{\mathcal{A}} + d_C + d_\tau^L), \quad K_\tau^R(C, \mathcal{A}) = (C \otimes \mathcal{A}, d_C + d_{\mathcal{A}} + d_\tau^R)$$

are acyclic;

- (c) the PBW spectral sequence is complete, separated, and Mittag–Leffler on finite windows, so the associated graded acyclicity lifts to the completed bar complex.

The chiral algebra $\mathcal{A}$ is *chirally Koszul* if the canonical bar twisting datum $(\mathcal{A}, \bar{B}^{\text{ch}}(\mathcal{A}), \tau_o, F_\bullet^{\text{PBW}})$ is Koszul.

Theorem 3.2 (Conditional chart criteria for chiral Koszulness). *Let $\mathcal{A}$ be a positive-energy chiral algebra on X with finite PBW windows, PBW-flat associated graded, and a complete separated Mittag–Leffler PBW filtration. Conditions (i)–(iv), (vi)–(x), and (xii)–(xiii) below are bidirectionally equivalent under the Theorem-B ambient hypotheses for the chiral bar–cobar adjunction. Condition (xi) is equivalent to them for all chiral algebras whose FM dual complex is simplicial. Condition (v) is a proved one-way consequence on the Koszul locus, with a conditional converse stated in Theorem 3.5. Condition (xiv) is equivalent to the others for the affine Kac–Moody family and one-directional in general.*

Under the perfectness hypothesis on the ambient tangent complex, the Lagrangian criterion (xv) is also equivalent. $\mathcal{D}$ -module purity (xvi) implies (x); the converse is proved for affine Kac–Moody and open in general.

Core equivalences (homological algebra and factorization geometry):

- (i) **Chiral Koszulness.** *The twisted tensor products $K_\tau^L(\mathcal{A}, \bar{B}^{\text{ch}}(\mathcal{A}))$ and $K_\tau^R(\bar{B}^{\text{ch}}(\mathcal{A}), \mathcal{A})$ are acyclic (Definition 3.1).*
- (ii) **PBW E_2 -collapse.** *The PBW spectral sequence $E_1^{p,q} = H^{p+q}(\text{gr}_p^F \bar{B}^{\text{ch}}(\mathcal{A})) \Rightarrow H^{p+q}(\bar{B}^{\text{ch}}(\mathcal{A}))$ collapses at E_2 : all differentials $d_r = 0$ for $r \geq 2$.*
- (iii) **A_∞ -formality of bar cohomology.** *The minimal A_∞ -model of $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$ is formal: the transferred higher products satisfy $m_n = 0$ for all $n \geq 3$.*

- (iv) **Shadow-tower formality.** The genus-zero obstruction classes of the shadow tower vanish: $o_{r+1}^{(o)}(\mathcal{A}) = 0$ for all $r \geq 2$. Equivalently, the transferred L_∞ -brackets $\ell_{r+1}^{(o)}$ of the modular convolution algebra vanish at genus zero for all $r \geq 2$.
- (v) **E_2 -formality of chiral Hochschild cohomology.** $\text{ChirHoch}^*(\mathcal{A})$ is concentrated in cohomological degrees $\{0, 1, 2\}$, carries a polynomial Hilbert series $P_{\mathcal{A}}(t) = \dim Z(\mathcal{A}) + \dim \text{ChirHoch}^1(\mathcal{A}) \cdot t + \dim Z(\mathcal{A}^!) \cdot t^2$, and is formal as an E_2 -algebra.
- (vi) **Curve independence.** The genus-zero factorization homology $H^k(\int_{\mathbb{P}^1} \mathcal{A})$ is independent of the choice of smooth projective curve: replacing $\mathbb{P}^1$ by any smooth proper curve X' of genus zero gives the same cohomology.
- (vii) **PBW universality.** The PBW filtration on $\mathcal{A}$ deforms flatly: $\text{gr}^F \mathcal{A} \cong \mathcal{A}_0$ for the one-parameter deformation, and the associated graded is quadratic-Koszul in the classical sense. Sufficient condition: $\mathcal{A}$ is freely strongly generated.
- (viii) **Barr–Beck–Lurie monadicity.** The comparison functor Φ for the bar–cobar adjunction $\bar{B}^{\text{ch}} \dashv \Omega^{\text{ch}}$ is an equivalence on the fiber over $\bar{B}^{\text{ch}}(\mathcal{A})$.
- (ix) **Factorization homology concentration.** $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$. Under the uniform-weight hypothesis, for every smooth projective curve Σ_g of genus $g \geq 0$, $H^k(\int_{\Sigma_g} \mathcal{A}) = 0$ for $k \neq 0$, and the surviving scalar class at $g \geq 1$ is $\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT).
- (x) **FM boundary acyclicity.** For every Fulton–MacPherson boundary stratum $S_T \cong \prod_{v \in V(T)} \bar{C}_{|v|}(X)$ indexed by a rooted tree T , the restriction $i_{S_T}^! \bar{B}_n^{\text{ch}}(\mathcal{A})$ is acyclic outside degree 0.

Tropical and representation-theoretic:

- (xi) **Tropical Koszulness.** The tropicalization of the bar complex $\text{trop}(\bar{B}^{\text{ch}}(\mathcal{A}))$, obtained by passing to the dual complex of the FM compactification and retaining only the combinatorial (tree-level) data, satisfies tropical E_2 -collapse: the tropical spectral sequence has $d_r^{\text{trop}} = 0$ for $r \geq 2$.
- (xii) **Ext diagonal vanishing.** $\text{Ext}_{\mathcal{A}}^{p,q}(\omega_X, \omega_X) = 0$ for $p \neq q$, where the bigrading comes from the PBW filtration on the bar resolution.
- (xiii) **Kac–Shapovalov determinant nonvanishing.** The Shapovalov determinant $\det G_h \neq 0$ in the bar-relevant range: the PBW-to-bar comparison map is injective at every bar-relevant conformal weight h .

Poisson-geometric (affine family):

- (xiv) **Sklyanin** $H^2 = 0$. For $\mathcal{A} = V_k(\mathfrak{g})$ with $\mathfrak{g}$ semisimple, the second Poisson cohomology of the Sklyanin bracket $\{-, -\}_{\text{STS}}$ on $(\mathfrak{g}^!)^*$ vanishes: $H_\pi^2((\mathfrak{g}^!)^*, \{-, -\}_{\text{STS}}) = 0$. Equivalently, the Sklyanin–Poisson structure determined by the classical r -matrix $r(z) = k\Omega/z$ admits no nontrivial infinitesimal deformations.

Conditional and partial:

- (xv) **Lagrangian criterion** (conditional on perfectness). $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^!}$ are transverse Lagrangians in the (-1) -shifted symplectic deformation space $\mathcal{M}_{\text{comp}}$.
- (xvi) **$\mathcal{D}$ -module purity** (one-directional). Each $\bar{B}_n^{\text{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module, with characteristic variety aligned to FM boundary strata. Implies (x). Converse proved for affine Kac–Moody; open in general.

Remark 3.3 (Numbering convention). The numbering (i)–(xiv) groups conditions by mathematical kinship. The parent monograph [23] numbers twelve conditions (i)–(xii) in a slightly different order; the present list expands by introducing shadow-tower formality (iv), E_2 -formality (v), curve independence (vi), PBW universality (vii), and tropical Koszulness (xi) as named conditions, and by separating the Sklyanin criterion (xiv) from the Ext diagonal vanishing (xii). The Lagrangian criterion (xv) and $\mathcal{D}$ -module purity (xvi) are kept outside the numbered list because their equivalence class is incomplete.

Remark 3.4 (Logical strength of the chart list). The headline count *fourteen* aggregates conditions of three logical strengths; the list does not assert that all sixteen labeled items (i)–(xvi) are unconditional biconditionals.

Unconditional biconditionals (eleven total): (i) Koszul, (ii) PBW E_2 , (iii) A_∞ -formality, (iv) shadow-tower formality, (vi) curve independence, (vii) PBW universality, (viii) Barr–Beck–Lurie monadicity, (ix) factorization-homology concentration, (x) FM boundary acyclicity, (xii) Ext diagonal, (xiii) Kac–Shapovalov determinant.

One-directional implications (two total): (v) E_2 -formality of ChirHoch^* : direction (i) $\Rightarrow$ (v) proved, converse conditional in Theorem 3.5. (xvi) $\mathcal{D}$ -module purity: (xvi) $\Rightarrow$ (x) proved, converse for affine Kac–Moody only.

Conditional on hypothesis (two total): (xi) tropical Koszulness, equivalent when the FM dual complex is simplicial (true for smooth curves). (xv) Lagrangian criterion, equivalent under perfectness and non-degeneracy of the ambient tangent complex.

Family-restricted (one total): (xiv) Sklyanin $H^2 = 0$, biconditional for the affine Kac–Moody family, one-directional ($\Rightarrow$ (ii)) in general.

Count: $11 + 2 + 2 + 1 = 16$ labeled items. The headline *fourteen* corresponds to (i)–(xiv); the conditional pair (xv), (xvi) is kept outside.

Theorem 3.5 (Conditional Hochschild converse). *Let $\mathcal{A}$ be a positive-energy chiral algebra on X with PBW filtration. Suppose condition (v) holds in its full form: $\text{ChirHoch}^*(\mathcal{A})$ is concentrated in cohomological degrees $\{0, 1, 2\}$, carries the polynomial Hilbert series $P_{\mathcal{A}}(t) = \dim Z(\mathcal{A}) + \dim \text{ChirHoch}^1(\mathcal{A}) \cdot t + \dim Z(\mathcal{A}^1) \cdot t^2$, and is formal as an E_2 -algebra. Assume in addition:*

- (a) *a two-colour chiral Higher Deligne comparison identifies $\text{ChirHoch}^*(\mathcal{A})$ with the derived E_2 -chiral centre of $\mathcal{A}$ at the chosen associator;*
- (b) *the Koszul dual of $\text{SC}^{\text{ch}, \text{top}}$ admits a contracting homotopy on the finite PBW component under consideration;*
- (c) *the chiral cap product detects the bar cohomology operations faithfully.*

Then condition (v) implies chiral Koszulness (i).

Proof. Four steps.

Step 1: Chiral Deligne–Tamarkin identification. Under the assumed two-colour chiral Higher Deligne comparison, the E_2 -algebra structure on $\text{ChirHoch}^*(\mathcal{A})$ is identified with the derived E_2 -chiral centre $\mathcal{Z}_{\text{ch}, E_2}^{\text{der}}(\mathcal{A})$ via

$$\text{ChirHoch}^*(\mathcal{A}) \simeq \text{RHom}_{E_2^{\text{ch-mod}}(\mathcal{A}, \mathcal{A})},$$

where the right-hand side is computed in the factorization ambient on $\mathbb{P}^1$. This identification is associator-dependent; fix a Drinfeld associator Φ (e.g. Φ_{KZ}).

Step 2: Cobar of formal E_2 -algebra. The E_2 -formality hypothesis on $\text{ChirHoch}^*(\mathcal{A})$ gives a strict E_2 -algebra model. Kontsevich–Tamarkin operadic Koszul duality between E_2 and its shifted dual $E_2\{1\}$ produces an E_2 -cobar $\Omega_{E_2}(\text{ChirHoch}^*(\mathcal{A}))$ that computes $\text{RHom}_{E_2\text{-mod}}(k, \mathcal{A})$ in the chiral ambient. The polynomial Hilbert series hypothesis ($\dim = \dim Z(\mathcal{A}) + \dim \text{ChirHoch}^1 \cdot t + \dim Z(\mathcal{A}^1) \cdot t^2$,

no higher) forces the brace operations br_n on $\text{ChirHoch}^*(\mathcal{A})$ to vanish for all $n \geq 3$: any nonzero br_n would contribute a class in degree ≥ 3 by arity-counting on $\overline{C}_n(\mathbb{C})$ -chains, contradicting concentration in $\{0, 1, 2\}$.

Step 3: Transfer to A_∞ -operations on bar cohomology. The bar coalgebra $\bar{B}^{\text{ch}}(\mathcal{A})$ and its dual $R\text{Hom}(\bar{B}^{\text{ch}}(\mathcal{A}), k) = C_{\text{ch}}^*(\mathcal{A}, k)$ are related to $\text{ChirHoch}^*(\mathcal{A})$ by the chiral cap product. Under the chiral Deligne identification of Step 1, the transferred A_∞ -operations m_n on $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$ are images of the brace operations br_n under the Koszul dual pairing. Brace vanishing for $n \geq 3$ (Step 2) forces $m_n = 0$ for all $n \geq 3$. This is condition (iii), A_∞ -formality of bar cohomology.

Step 4: Apply the core criterion. By the core circuit of Theorem 3.2, (iii) $\Leftrightarrow$ (i): A_∞ -formality of bar cohomology is equivalent to chiral Koszulness. Hence (v) $\Rightarrow$ (iii) $\Rightarrow$ (i) under the three comparison hypotheses.

The proof requires the associator choice Φ of Step 1, so the converse is chart-dependent. Associator-independent two-colour formality is a separate theorem-level obligation. $\square$

Remark 3.6 (Dependence on chiral Higher Deligne). The theorem isolates the exact hypotheses needed for the converse: two-colour chiral Higher Deligne, a finite-PBW contracting homotopy for the Koszul dual Swiss-cheese complex, and faithful cap-product detection. The associator-independent formulation remains open.

Remark 3.7 (The four objects). The four objects of chiral Koszul duality must not be conflated:

- (i) $\bar{B}^{\text{ch}}(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$: the bar *coalgebra*, with deconcatenation coproduct and bar differential.
- (ii) $\mathcal{A}^i = H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$: the dual *coalgebra*, Koszul cohomology of the bar complex.
- (iii) $\mathcal{A}^! = ((\mathcal{A}^i)^\vee)$: the Koszul dual *algebra*, obtained by Verdier duality.
- (iv) $\mathcal{Z}_{\text{ch}}^{\text{der}}(\mathcal{A}) = R\text{Hom}_{\mathcal{A} \otimes \mathcal{A}^{\text{op}}}(\mathcal{A}, \mathcal{A})$: the derived chiral center, computed using the bar complex as a resolution.

The bar complex is the E_1 engine; $\Omega(\bar{B}^{\text{ch}}(\mathcal{A})) = \mathcal{A}$ is *inversion*, not Koszul duality. The Koszul dual comes from Verdier duality. The derived center comes from chiral Hochschild cochains.

4. THE CIRCUIT OF IMPLICATIONS

The proof of Theorem 3.2 is organized as a directed graph of implications. The central structure is the *core circuit*, a cycle of biconditional implications among five conditions:

$$(i) \Leftrightarrow (ii) \Leftrightarrow (iii) \Leftrightarrow (viii) \Leftrightarrow (ix).$$

Each of the remaining conditions is shown equivalent to a designated node of this core by a satellite implication. The full graph has the following shape.

On the Koszul locus, the bar functor detects quasi-isomorphisms (the bar-cobar counit inverts them), giving conservativity. The bar-cobar adjunction is a Quillen equivalence on the Koszul locus [23,

§5]; a Quillen equivalence of stable model categories satisfies the monadic descent conditions of [19, Prop. 4.7.3.16]. Hence BBL monadicity holds if and only if the chiral algebra is Koszul.

(i) $\Leftrightarrow$ (ix): *Koszulness and factorization homology concentration.* The bar complex computes factorization homology: $\bar{B}_n^{\text{ch}}(\mathcal{A}) = (\int_X \mathcal{A})_n$. At genus zero, E_2 -collapse gives $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$: the Goodwillie filtration has associated graded given by bar contributions, each concentrated in degree 0 by PBW collapse. Under the uniform-weight hypothesis, the same holds at every genus: the loop-order spectral sequence collapses on the modular Koszul locus, each vertex contribution remains in degree 0, and the surviving scalar is $\text{obs}_g(\mathcal{A}) = \kappa(\mathcal{A}) \cdot \lambda_g$ (UNIFORM-WEIGHT).

Conversely, concentration at genus zero specializes to $H^k(\int_{\mathbb{P}^1} \mathcal{A}) = 0$ for $k \neq 0$. This is computed by the bar complex: the geometric realization $\int_{\mathbb{P}^1} \mathcal{A} \simeq |\bar{B}^{\text{ch}}(\mathcal{A})|$. Concentration in degree 0 forces the bar spectral sequence to collapse, which by the first equivalence is Koszulness.

4.2. Satellite equivalences from the core.

(ii) $\Leftrightarrow$ (iv): *PBW collapse and shadow-tower formality.* The shadow obstruction tower $\{o_{r+1}(\mathcal{A})\}_{r \geq 2}$ has two projections: the genus-zero slice $o_{r+1}^{(0)}(\mathcal{A})$ and the all-genera projection. Shadow-tower formality (vanishing of the genus-zero slice) is identified with the transferred L_∞ -brackets $\ell_{r+1}^{(0)}$ of the modular convolution algebra at genus zero. The HPL transfer identifies these with the PBW spectral sequence differentials. Hence shadow-tower formality is equivalent to E_2 -collapse.

The all-genera shadow tower, by contrast, can be nontrivial even for Koszul algebras: this is the shadow depth classification G/L/C/M. Shadow-tower formality is the genus-zero condition; shadow depth is the all-genera condition. They are independent invariants.

(i) $\Rightarrow$ (v): *Koszulness implies E_2 -formality of ChirHoch.* The bar–cobar quasi-isomorphism $\Omega^{\text{ch}}(\bar{B}^{\text{ch}}(\mathcal{A})) \xrightarrow{\sim} \mathcal{A}$ provides a free resolution of $\mathcal{A}$. Computing $\text{ChirHoch}^*(\mathcal{A})$ through this resolution, the bar–cobar filtration forces diagonal concentration. The polynomial structure in degrees $\{0, 1, 2\}$ follows: degree 0 is the center, degree 1 is first-order deformations, degree 2 is obstructions. The E_2 -formality of $\text{ChirHoch}^*(\mathcal{A})$ uses the formality of the local configuration spaces $\bar{C}_n(\mathbb{C})$.

This is a one-way consequence: E_2 -formality of ChirHoch^* does not force the underlying graded-commutative algebra to be free, so the converse from (v) to bar–cobar inversion is not established.

(ii) $\Leftrightarrow$ (vi): *PBW collapse and curve independence.* Curve independence at genus zero follows from PBW collapse because the collapsed spectral sequence depends only on the associated graded $R_{\mathcal{A}}$, which is a Poisson vertex algebra independent of the global geometry of X . The FM compactification $\bar{C}_n(X)$ depends on X , but the E_2 -page $H^{p+q}(\text{gr}_p^F \bar{B}^{\text{ch}}(\mathcal{A}))$ is computed from $R_{\mathcal{A}}$ by Chevalley–Eilenberg methods that see only the local OPE data.

Conversely, if factorization homology is curve-independent at genus zero, the bar complex cannot have differentials sensitive to global geometry. The PBW spectral sequence differentials d_r for $r \geq 2$ involve integration over FM strata of codimension r ; these strata encode collisions of $r + 1$ points and are sensitive to the curve’s geometry at the r -fold collision scale. Curve independence forces these to vanish, giving E_2 -collapse.

(iii) $\Leftrightarrow$ (vii): *A_∞ -formality and PBW universality.* PBW universality states that the PBW filtration deforms flatly with classically Koszul associated graded. If $\mathcal{A}$ is freely strongly generated (the generators form a free $\mathcal{D}_X$ -module), the associated graded is a free polynomial PVA, which is quadratic-Koszul. The flat deformation property then implies that the bar spectral sequence for $\mathcal{A}$ is a deformation of the (collapsed) spectral sequence for $\mathcal{A}_0$. By semicontinuity, the collapsed page persists, giving E_2 -collapse and hence A_∞ -formality.

Conversely, A_∞ -formality of bar cohomology means the transferred operations $m_n = 0$ for $n \geq 3$; this is the linearization condition that PBW universality encodes.

(i) $\Leftrightarrow$ (x): *Koszulness and FM boundary acyclicity.* Each FM boundary stratum S_T is indexed by a rooted tree T . The residue component $d_{\text{res}}|_{S_T}$ of the bar differential restricts to the tensor product of lower-degree bar complexes at the vertices of T . E_2 -collapse at each vertex gives acyclicity at S_T : $H^k(i_{S_T}^! \bar{B}_n^{\text{ch}}(\mathcal{A})) = 0$ for $k \neq 0$.

Conversely, acyclicity at the deepest strata (binary collision divisors $S = D_{\{i,j\}} \cong X \times \bar{C}_{n-1}(X)$) forces every OPE residue to be exact in the PBW-graded sense, which is the condition for E_2 -collapse. Iterating over all strata closes the equivalence.

(x) $\Leftrightarrow$ (xi): *FM boundary acyclicity and tropical Koszulness.* The tropicalization of the bar complex retains only the combinatorial data visible in the dual complex of the FM compactification: the rooted trees indexing boundary strata, the face maps between strata, and the residues of the bar differential along each stratum. Tropical E_2 -collapse is the assertion that these combinatorial residues produce no off-diagonal cohomology.

FM boundary acyclicity (the analytic statement) implies tropical Koszulness (the combinatorial statement) by restriction. The converse holds when the dual complex is simplicial (the standard case for smooth curves): the combinatorial data determines the analytic behavior up to homotopy equivalence, so tropical acyclicity forces analytic acyclicity.

(i) $\Leftrightarrow$ (xii): *Koszulness and Ext diagonal vanishing.* E_2 -collapse places $H^*(\bar{B}^{\text{ch}}(\mathcal{A}))$ on the PBW diagonal $p = q$. The bar complex computes $\text{Ext}_{\mathcal{A}}^{p,q}(\omega_X, \omega_X)$, so diagonal concentration of bar cohomology is Ext diagonal vanishing.

Conversely, Ext diagonal vanishing means the abutment of the PBW spectral sequence is diagonal. Any nonzero differential d_r for $r \geq 2$ would produce off-diagonal classes, contradicting diagonal concentration. Hence $d_r = 0$ for all $r \geq 2$.

(i) $\Leftrightarrow$ (xiii): *Koszulness and Kac–Shapovalov determinant.* If $\mathcal{A}$ is Koszul, the PBW filtration is strict on the vacuum module: $\text{gr}^F M_0(\mathcal{A}) \cong \text{Sym}^{\text{ch}}(V)$. Strictness implies non-degeneracy of the Shapovalov form G_h in the bar-relevant range.

Conversely, non-degeneracy of G_h makes the PBW-to-bar comparison injective at every bar-relevant weight. Suppose $d_r \neq 0$ for some $r \geq 2$; then there exists a nonzero class in $E_r^{p,q}$ with nonzero differential, whose lift to the bar complex contradicts the acyclicity of $\text{Sym}(V)$ as a Koszul complex. Hence $d_r = 0$ for all $r \geq 2$.

(xiv) $\Rightarrow$ (ii): *Sklyanin $H^2 = 0$ implies PBW collapse (affine family).* For $\mathcal{A} = V_k(\mathfrak{g})$ with $\mathfrak{g}$ semisimple, the Koszul dual coalgebra $\bar{B}^{\text{ch}}(\mathcal{A})^!$ carries the Poisson–Lie group $(\mathfrak{g}^!)^*$ with the Sklyanin–Poisson bracket determined by the classical r -matrix $r(z) = k\Omega/z$ via the Semenov–Tian–Shansky construction. The second Poisson cohomology H_π^2 controls infinitesimal deformations of this bracket.

The vanishing $H_\pi^2 = 0$ is established by Whitehead’s second lemma (for $\mathfrak{g}$ semisimple, $H^2(\mathfrak{g}; M) = 0$ for finite-dimensional modules M): the Poisson cochain complex $(\mathfrak{X}^\bullet, \delta_\pi)$ is the associated graded of the bar differential under PBW filtration, so vanishing of H_π^2 is the associated-graded avatar of E_2 -collapse. The proof: five independent derivation paths (direct kernel/image, Chevalley–Eilenberg comparison, polynomial-degree spectral sequence, Whitehead reduction, and numerical from the compute engine) confirm $H_\pi^0 = \mathbb{C}[C_2]$ (Casimir ring), $H_\pi^1 = H_\pi^2 = H_\pi^3 = 0$.

This is proved as a biconditional for the affine Kac–Moody family. For general chiral algebras, the Sklyanin construction does not apply (it requires a Lie-algebraic r -matrix), so the equivalence is restricted to this family.

(xv): *Lagrangian criterion, conditionally.* Under the perfectness and non-degeneracy hypotheses on the ambient tangent complex: on the Koszul locus, the complementarity splitting

$$\mathbf{Q}_g(\mathcal{A}) \oplus \mathbf{Q}_g(\mathcal{A}^\perp) \simeq H^*(\overline{\mathcal{M}}_g, \mathcal{Z}_{\mathcal{A}})$$

identifies $\mathcal{M}_{\mathcal{A}}$ and $\mathcal{M}_{\mathcal{A}^\perp}$ as complementary subspaces of $\mathcal{M}_{\text{comp}}$. The (-1) -shifted symplectic structure makes each isotropic; complementarity gives maximal dimension, hence Lagrangian. Conversely, transverse Lagrangian intersection in a (-1) -shifted symplectic space has expected dimension zero; its discreteness is acyclicity of the twisted tensor product, which is Koszulness.

(xvi) $\Rightarrow$ (x): *$\mathcal{D}$ -module purity implies FM boundary acyclicity.* Suppose each $\bar{B}_n^{\text{ch}}(\mathcal{A})$ is pure of weight n as a mixed Hodge module. By Saito's theory, the weight filtration on a pure Hodge module splits up to semisimplification; purity plus the alignment of the characteristic variety to FM strata forces acyclicity outside degree 0 on every stratum. The converse (Koszulness implies purity) is proved for the affine Kac–Moody family, where the bar complex is identified with a geometric complex whose purity follows from the Riemann hypothesis for weight filtrations.

5. SCALAR SATURATION

5.1. **The modular characteristic.** Let $\mathcal{A}$ be a chiral algebra on X . The universal Maurer–Cartan element $\Theta_{\mathcal{A}} := D_{\mathcal{A}} - d_o$ of the bar differential lives in the modular convolution dg Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ and satisfies

$$d \Theta_{\mathcal{A}} + \frac{1}{2} [\Theta_{\mathcal{A}}, \Theta_{\mathcal{A}}] = 0.$$

At each degree $r \geq 2$, $\Theta_{\mathcal{A}}$ projects to a shadow invariant $S_r(\mathcal{A})$.

The E_1 ordered bar complex $\bar{B}^{\text{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is the primitive object: it carries the full R -matrix $r_{\mathcal{A}}(z) = \text{Res}_{o,2}^{\text{coll}}(\Theta_{\mathcal{A}})$. The symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$ is the Σ_n -coinvariant quotient: a lossy projection that discards the matrix-valued data and retains only the scalar

$$\kappa(\mathcal{A}) = S_2(\mathcal{A}),$$

the *modular characteristic*: the single number that survives averaging.

Remark 5.1 (Family-specific formulas). The modular characteristic is family-specific; writing it from the canonical landscape table prevents convention drift. The verified formulas are:

Family	$\kappa(\mathcal{A})$	Sanity checks
Heisenberg $\mathcal{H}_k$	k	$k = 0 \rightarrow 0; k = 1 \rightarrow 1$
Affine KM $V_k(\mathfrak{g})$	$\frac{\dim(\mathfrak{g})(k + h^\vee)}{2h^\vee}$	$k = 0 \rightarrow \frac{\dim(\mathfrak{g})}{2}; k = -h^\vee \rightarrow 0$
Virasoro Vir_c	$c/2$	$c = 0 \rightarrow 0; c = 13 \rightarrow 13/2$
$\mathcal{W}_N$ (principal)	$c \cdot (H_N - 1)$	$N = 2 \rightarrow c/2 = \kappa^{\text{Vir}}; N = 3 \rightarrow 5c/6$

where $H_N = \sum_{j=1}^N 1/j$ is the N -th harmonic number.

5.2. The saturation theorem.

Theorem 5.2 (Scalar saturation on finite PBW charts). *Let $\mathcal{A}_k$ be a one-parameter algebraic family of chiral algebras with rational OPE coefficients, parametrized by a level k . Assume a finite PBW–Artin chart, scalar-diagonal Zamolodchikov metric, and a one-dimensional degree-two deformation space. Then the scalar projection of the degree-two universal MC element is determined by the single scalar $\kappa(\mathcal{A}_k)$:*

$$\text{av}\left(\Theta_{\mathcal{A}_k}^{(2)}\right) = \kappa(\mathcal{A}_k) \cdot \eta \otimes \Lambda,$$

where $\eta = d \log(z_1 - z_2)$ is the Arnold propagator and Λ is the universal quadratic Casimir element.

Proof sketch. The degree-two component of $\Theta_{\mathcal{A}}$ in the modular convolution algebra takes values in $H^2(\mathfrak{g}_{\mathcal{A}}^{\text{mod}}; \text{Sym}^q V)$ for appropriate q . Whitehead's second lemma gives vanishing of this cohomology for reductive Lie algebras at generic level: the space of degree-two MC deformations is one-dimensional, spanned by $\eta \otimes \Lambda$. The coefficient is determined by the Σ_n -coinvariant projection: $\text{av}(\Theta_{\mathcal{A}}^{(2)}) = \kappa(\mathcal{A}) \cdot [\lambda_1]$, which fixes the coefficient as κ .

The argument has three steps:

- (i) **Whitehead reduction.** For each standard family, the Lie algebra cohomology $H^2(\mathfrak{g}; \text{Sym}^q V) = 0$ at generic level (the reductive hypothesis is used here). This gives a one-dimensional deformation space at degree two.
- (ii) **Algebraic semicontinuity.** The exceptional locus where Whitehead reduction fails is Zariski-closed in the level parameter k . The theorem is asserted on the complement of this exceptional locus and on the named standard rows where the finite-PBW computation has been carried out.
- (iii) **Coefficient identification.** The averaging map $\text{av}: \mathfrak{g}_{\mathcal{A}}^{E_1} \rightarrow \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ projects $\Theta_{\mathcal{A}}^{(2)}$ to $\kappa(\mathcal{A}) \cdot [\lambda_1]$. Since av is the Σ_n -coinvariant projection and $\eta \otimes \Lambda$ is the unique Σ_2 -invariant degree-two element (up to scalar), the coefficient is κ .

For non-abelian affine KM, the averaging identity involves the Sugawara shift: $\text{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa(V_k(\mathfrak{g}))$ where $\text{av}(r(z)) = k \cdot \dim(\mathfrak{g})/(2h^\vee)$ is the double-pole channel and $\dim(\mathfrak{g})/2$ is the simple-pole self-contraction through the adjoint Casimir. $\square$

Remark 5.3 (Scope). Scalar saturation at degree 2 is proved on the scalar-diagonal standard rows and on finite PBW-Artin components satisfying the hypotheses of Theorem 5.2. For multi-weight algebras at genus $g \geq 2$, the scalar projection saturates while the full MC element receives cross-channel corrections. The stronger identity $\Theta_{\mathcal{A}}^{\text{min}} = \kappa \cdot \eta \otimes \Lambda$ is proved under the uniform-weight hypothesis. At $g \geq 2$ it receives cross-channel corrections for multi-weight families.

5.3. From scalar saturation to Koszulness. Scalar saturation establishes that κ controls the MC element at degree 2. The question is what controls it at *all* degrees. The answer is the Koszul condition.

Proposition 5.4 (Koszulness controls the full tower). *If $\mathcal{A}$ is chirally Koszul, then the universal MC element $\Theta_{\mathcal{A}}$ is determined at every degree by its genus-zero projection. Concretely:*

- (i) *The shadow invariants $S_r(\mathcal{A})$ for $r \geq 3$ are determined by the A_∞ -formality data of bar cohomology.*
- (ii) *Under the uniform-weight hypothesis, $\Theta_{\mathcal{A}}$ is determined by $\kappa(\mathcal{A})$ and the shadow depth class.*
- (iii) *For class G algebras (Heisenberg), $S_r = 0$ for $r \geq 3$; the MC element is entirely controlled by κ .*
- (iv) *For class M algebras (Virasoro), $S_r \neq 0$ for infinitely many r , but each S_r is determined by $\kappa = c/2$ and $S_4 = 10/[c(5c + 22)]$.*

Proof sketch. A_∞ -formality (condition (iii) of Theorem 3.2) means $m_n = 0$ for $n \geq 3$; the bar cohomology algebra has no higher operations. The MC moduli of a formal A_∞ -algebra is discrete; the shadow invariants S_r are the Taylor coefficients of the unique MC solution in the completion of the convolution algebra. Each S_r is determined recursively from $\kappa = S_2$ and the structure constants of the algebra.

For class G, all $S_r = 0$ for $r \geq 3$ by direct computation (the bar complex is abelian). For class M, the recursion is controlled by the Riccati equation $\Delta = 8\kappa S_4$, where $\Delta = 0$ characterizes class G. The seed S_4 determines S_r for all r via the shadow algebra Lie bracket; for Virasoro, $S_4 = 10/[c(5c + 22)] \sim 2/c^2$ at large c . $\square$

Remark 5.5 (The critical discriminant). The discriminant $\Delta = 8\kappa \cdot S_4$ is linear in κ (not quadratic). $\Delta = 0$ if and only if the shadow tower is finite. For $\kappa \neq 0$, finiteness is equivalent to $S_4 = 0$, which characterizes class G. For $\kappa = 0$ (critical level), $\Delta = 0$ regardless of S_4 .

5.4. The E_1 structure. Scalar saturation is a statement about the Σ_n -coinvariant quotient. The full E_1 ordered bar complex carries strictly more data: the matrix-valued classical r -matrix $r_{\mathcal{A}}(z) \in \text{End}_{\mathcal{A}}(2)[[z^{-1}]]$, of which κ is only the scalar shadow $\kappa = \text{av}(r_{\mathcal{A}}(z))$.

The r -matrix satisfies the classical Yang–Baxter equation $[r_{12}, r_{13}] + [r_{12}, r_{23}] + [r_{13}, r_{23}] = 0$ and determines the Yangian structure on the ordered bar cohomology. It absorbs one pole from the OPE via the logarithmic propagator: for Heisenberg, the OPE has a double pole and the r -matrix has a simple pole, $r^{\text{Heis}}(z) = k/z$. For Virasoro, the OPE has a quartic pole and the r -matrix has a cubic pole, $r^{\text{Vir}}(z) = (c/2)/z^3 + 2T/z$. For affine KM in the trace-form convention, $r^{\text{KM}}(z) = k\Omega/z$, where Ω is the inverse Killing form Casimir and the level prefix k is mandatory: at $k = 0$ the r -matrix vanishes (abelian limit).

The averaging map $\text{av}: \mathfrak{g}_{\mathcal{A}}^{E_1} \rightarrow \mathfrak{g}_{\mathcal{A}}^{\text{mod}}$ is the Σ_n -coinvariant projection. It is lossy: the full r -matrix determines κ , but κ does not determine the r -matrix. The five theorems of the parent monograph extract the Σ_n -invariant content; the present paper works at this invariant level, where scalar saturation gives full control at degree two and Koszulness extends control to all degrees.

6. THE STANDARD LANDSCAPE

The fourteen characterizations of Theorem 3.2 are verified on the standard generic rows of the landscape. The verification proceeds family by family, each through a criterion adapted to the structure of the algebra.

Theorem 6.1 (Standard generic families). *The following chiral algebras are chirally Koszul: Heisenberg $\mathcal{H}_k$ at all levels k ; lattice algebras V_L for all even lattices L ; affine Kac–Moody $V_k(\mathfrak{g})$ for semisimple $\mathfrak{g}$ at generic non-critical levels; Virasoro Vir_c at generic central charge; the principal $\mathcal{W}$ -algebra $\mathcal{W}_N$ at all N and generic level; the $\beta\gamma$ system at standard conformal weight; and the Bershadsky–Polyakov algebra $\mathcal{W}_k^{(2)}(\mathfrak{sl}_3)$ at generic level.*

The proof is the content of this section. Table 1 records the family, the Koszul equivalence that proves it, the shadow class, the modular characteristic, and the Koszul complementarity conductor.

Family	Koszul equivalence	Class	κ	K
$\mathcal{H}_k$	(ii) PBW, trivial	G	k	0
Lattice V_L	(ii) PBW, lattice	G	$\text{rk}(L)/2$	0
$V_k(\mathfrak{g})$	(xiii) Kac–Shapovalov	L	$\frac{\dim(\mathfrak{g})(k+h^\vee)}{2h^\vee}$	0
Vir_c	(iv) shadow formality	M	$c/2$	13
$\mathcal{W}_N$	(xiii) via DS from KM	M	$c(H_N - 1)$	family-dep.
$\beta\gamma$	(iii) $\mathcal{A}_\infty$ -formality	C	$6\lambda^2 - 6\lambda + 1$	0
$\mathcal{W}_k^{(2)}(\mathfrak{sl}_3)$	(xii) Ext diag.	M	$c(k)/6$	98/3

TABLE 1. The standard landscape and Koszulness. $H_N = \sum_{j=1}^N 1/j$. The column K denotes the Koszul complementarity conductor $K = \kappa(\mathcal{A}) + \kappa(\mathcal{A}^\dagger)$.

6.1. Heisenberg. The Heisenberg algebra $\mathcal{H}_k$ has a single generator J with OPE $J(z)J(w) \sim k/(z-w)^2$, so the bar complex $\bar{B}^{\text{ch}}(\mathcal{H}_k)$ is the desuspended tensor coalgebra on a one-dimensional augmentation ideal. The PBW spectral sequence has a single nonzero column at $p = 1$, so E_2 -collapse is vacuous: there are no possible higher differentials. This is equivalence (ii) of Theorem 3.2, in its trivial form. The modular characteristic is $\kappa(\mathcal{H}_k) = k$, and the shadow tower terminates at $r = 2$: class G. The Koszul dual is $\mathcal{H}_k^! = \text{Sym}^{\text{ch}}(V^*)$ (the chiral symmetric algebra on the linear dual), with $\kappa(\mathcal{H}_k^!) = -k$, giving $K = k + (-k) = 0$.

6.2. Lattice algebras. The lattice vertex algebra V_L associated to an even lattice L of rank r is an extension of the Heisenberg algebra by vertex operators e^α for $\alpha \in L$. The PBW filtration on V_L has associated graded $\text{gr}^F V_L \cong \text{Sym}^{\text{ch}}(\mathfrak{h}) \otimes \mathbb{C}[L]$, where $\mathfrak{h} = L \otimes_{\mathbb{Z}} \mathbb{C}$. The lattice grading gives a decomposition of the bar complex into weight spaces, each of which reduces to a Heisenberg bar complex tensored with the group algebra. The PBW spectral sequence collapses at E_2 in each weight sector: equivalence (ii). The modular characteristic is $\kappa(V_L) = \text{rk}(L)/2$, and the shadow class is G.

6.3. Affine Kac–Moody. For $\mathcal{A} = V_k(\mathfrak{g})$ with $\mathfrak{g}$ semisimple and $k \neq -h^\vee$, the Koszulness proof goes through the Kac–Shapovalov criterion (xiii). The Shapovalov form G_h on the vacuum Verma module $M_0(V_k(\mathfrak{g}))$ is nondegenerate for k outside the critical hyperplane: the Shapovalov determinant is a product over positive roots and PBW-relevant weights, and its zeros are confined to the critical level and the reducibility locus [20, Thm. II.4].

At generic non-critical level, $\det G_h \neq 0$ in the bar-relevant range: the PBW-to-bar comparison is injective, and the PBW spectral sequence collapses at E_2 by Theorem 3.2 (xiii) $\Leftrightarrow$ (ii). This is the chiral analogue of the Garland–Lepowsky concentration theorem [17]: the cohomology of the bar complex is concentrated on the PBW diagonal, the higher differentials vanish, and the associated graded computes the Koszul dual coalgebra.

The modular characteristic is $\kappa(V_k(\mathfrak{g})) = \dim(\mathfrak{g})(k + h^\vee)/(2h^\vee)$. Sanity checks: $k = 0$ gives $\kappa = \dim(\mathfrak{g})/2$ in the abelian limit: the double-pole vanishes and the simple-pole self-contraction persists. The critical value $k = -h^\vee$ gives $\kappa = 0$ (critical level, where this Shapovalov proof does not apply and the Feigin–Frenkel centre changes the ambient problem). The shadow class is L: the cubic shadow $S_3 \neq 0$ detects the Lie bracket, but $S_4 = 0$, so the tower terminates at degree 3. The Koszul conductor is $K = \kappa(V_k(\mathfrak{g})) + \kappa(V_{k'}(\mathfrak{g})) = 0$ where $k' = -2h^\vee - k$.

6.4. Virasoro. The Virasoro algebra Vir_c has OPE pole order 4 and is not quadratic in any classical sense. Its Koszulness is proved through shadow-tower formality (iv). The PBW spectral sequence for Vir_c has higher differentials that could a priori be nonzero; the proof that they vanish uses the recursive structure of the Maurer–Cartan element.

At genus zero, the shadow obstruction tower has $o_{r+1}^{(0)} = 0$ for all $r \geq 2$: the transferred L_∞ -brackets $\ell_{r+1}^{(0)}$ vanish because the Virasoro vacuum module satisfies Kac determinant nonvanishing at generic c . Shadow-tower formality at genus zero is equivalence (iv) $\Leftrightarrow$ (ii): the PBW spectral sequence collapses.

The modular characteristic is $\kappa(\text{Vir}_c) = c/2$, and the shadow class is M: $S_4 = 10/[c(\zeta c + 22)] \neq 0$ for generic c , so the all-genera tower is infinite. The Koszul dual is $\text{Vir}_c^! = \text{Vir}_{26-c}$, giving $K = c/2 + (26 - c)/2 = 13$. The self-dual point is $c = 13$ (not $c = 26$).

6.5. Principal $\mathcal{W}$ -algebras. The principal $\mathcal{W}$ -algebra $\mathcal{W}_N$ at level k is obtained from $V_k(\mathfrak{sl}_N)$ by Drinfeld–Sokolov reduction with respect to the principal nilpotent f_{prin} . The Koszulness of $\mathcal{W}_N$ is derived from the Koszulness of the parent affine KM algebra through the functoriality of DS reduction with respect to the bar complex.

DS reduction preserves the Shapovalov criterion: it sends the Kac–Shapovalov form on $V_k(\mathfrak{sl}_N)$ to the $\mathcal{W}$ -algebra Shapovalov form on $\mathcal{W}_N$, and nonvanishing of the former at generic level implies

nonvanishing of the latter. This gives Koszulness of $\mathcal{W}_N$ through equivalence (xiii). At $N = 2$, this recovers the Virasoro result through the DS path from $\mathfrak{sl}_2$.

The generators of $\mathcal{W}_N$ have conformal weights $\{2, 3, \dots, N\}$: that is $N - 1$ generators, with highest weight N . The modular characteristic is $\kappa(\mathcal{W}_N) = c \cdot (H_N - 1)$ where $H_N = \sum_{j=1}^N 1/j$. Sanity check at $N = 2$: $H_2 - 1 = 3/2 - 1 = 1/2$, so $\kappa(\mathcal{W}_2) = c/2$, matching $\kappa(\text{Vir}_c)$. At $N = 3$: $H_3 - 1 = 11/6 - 1 = 5/6$, so $\kappa(\mathcal{W}_3) = 5c/6$.

6.6. The $\beta\gamma$ system. The $\beta\gamma$ system at conformal weight λ has central charge $c_{\beta\gamma}(\lambda) = 2(6\lambda^2 - 6\lambda + 1)$. Its bar complex has a contact-term tower that terminates at degree 4 (shadow class C), so the transferred $\mathcal{A}_\infty$ -products m_n satisfy $m_n = 0$ for $n \geq 5$. But Koszulness requires $m_n = 0$ for $n \geq 3$.

The vanishing of m_3 and m_4 is proved by explicit computation: the genus-zero shadow obstructions $o_3^{(0)} = o_4^{(0)} = 0$ at generic conformal weight. This establishes $\mathcal{A}_\infty$ -formality, equivalence (iii). The $\beta\gamma$ system is the simplest family in class C, the class where the tower is finite but nontrivial ($S_3 \neq 0$, $S_4 \neq 0$, $S_r = 0$ for $r \geq 5$).

6.7. Bershadsky–Polyakov. The Bershadsky–Polyakov algebra $\mathcal{W}_k^{(2)}(\mathfrak{sl}_3)$ is the non-principal $\mathcal{W}$ -algebra obtained by DS reduction of $V_k(\mathfrak{sl}_3)$ with respect to the minimal nilpotent. It has generators at conformal weights $\{1, 3/2, 3/2, 2\}$ and central charge, following Feigin–Kac–Roan–Wakimoto:

$$c(\text{BP}_k) = 2 - \frac{24(k+1)^2}{k+3}.$$

The formula is singular at the critical level $k = -3 = -h^\vee(\mathfrak{sl}_3)$, where the Sugawara denominator $k + h^\vee$ of the parent $V_k(\mathfrak{sl}_3)$ vanishes; the self-dual point $k = -3$ of the involution $k \mapsto -k - 6$ coincides with this critical locus, and the value $\kappa(\text{BP}_{-3}) = 49/3$ quoted below is the *principal value* of the symmetric limit across the pole, not a regular function value. The polynomial identity $K_{\text{BP}}(k) := c(k) + c(-k - 6) \equiv 196$ holds in $\mathbb{Q}(k)$, and $c(k) - 98$ is an odd function of $(k + 3)$.

Koszulness is established through Ext diagonal vanishing (xii): the bigraded Ext groups $\text{Ext}^{p,q}(\omega, \omega)$ are concentrated on the diagonal $p = q$ at generic level. The proof uses the non-principal DS reduction from $V_k(\mathfrak{sl}_3)$: the DS functor sends the Kac–Shapovalov form of the parent to the Bershadsky–Polyakov Shapovalov form, and nonvanishing at generic level gives diagonal concentration.

The modular characteristic is $\kappa(\mathcal{W}_k^{(2)}) = c(k)/6$, with anomaly ratio $\rho = 1/6$. The Koszul complementarity conductor is $K_{\text{BP}} = \kappa(\mathcal{W}_k^{(2)}) + \kappa(\mathcal{W}_{k'}^{(2)}) = 98/3$, where $k' = -k - 6$ is the dual level. The self-dual level is $k = -3$, at which $\kappa = 49/3$.

7. THE ADMISSIBLE-LEVEL QUESTION

The standard landscape verification of §6 treats the universal enveloping VOA $V_k(\mathfrak{g})$ at generic level. At admissible levels $k = -h^\vee + p/q$ (coprime $p \geq h^\vee$, $q \geq 2$), the simple quotient $L_k(\mathfrak{g}) = V_k(\mathfrak{g})/I_k$ is a rational vertex algebra. The question is whether the simple quotient remains Koszul; the universal-algebra argument above does not settle that quotient.

7.1. Rank one: $\mathfrak{sl}_2$.

Proposition 7.1 ($\mathfrak{sl}_2$ at admissible level). *The simple quotient $L_k(\mathfrak{sl}_2)$ is chirally Koszul at every admissible level $k = -2 + p/q$ with $p \geq 2$, $q \geq 2$, $\gcd(p, q) = 1$.*

Proof sketch. The ideal $I_k \subset V_k(\mathfrak{sl}_2)$ is generated by a single null vector at conformal weight pq . Since $\dim \mathfrak{sl}_2 = 3$, the PBW spectral sequence for $L_k(\mathfrak{sl}_2)$ has at most two nonzero columns at each conformal weight: the Cartan direction h and the root directions e, f . The root generators are paired by $[e, f] = h$ on the E_1 page, and the Cartan generator survives as a single class in H^1 .

The null vector at weight pq introduces a relation in the bar complex, but this relation is confined to a single PBW column: it does not create off-diagonal classes in the E_2 page. The PBW spectral sequence collapses at E_2 because the dimensional constraint ($\dim \mathfrak{sl}_2 = 3$, hence at most 2 nonzero columns in the relevant range) leaves no room for higher differentials d_r with $r \geq 2$. $\square$

Remark 7.2 (Mechanism). The proof for $\mathfrak{sl}_2$ uses the small-rank accident: three generators produce at most two nonzero PBW columns at each weight, and a differential d_r for $r \geq 2$ needs at least $r + 1$ nonzero columns as domain and target. At rank one no domain-target pair remains. This is specific to $\mathfrak{sl}_2$; it fails at every higher rank.

7.2. Rank two and beyond: the obstruction.

Conjecture 7.3 (Admissible Koszul rank obstruction). *For every simple Lie algebra $\mathfrak{g}$ of rank $r \geq 2$ and every admissible level $k = p/q - h^\vee$ with denominator $q \geq 3$, the simple quotient $L_k(\mathfrak{g})$ has $\dim H^2(\bar{B}^{\text{ch}}(L_k(\mathfrak{g}))) = \text{rank}(\mathfrak{g})$. Then $L_k(\mathfrak{g})$ is not chirally Koszul.*

The mechanism is the abelian Cartan subalgebra $\mathfrak{h} \subset \mathfrak{g}$. On the E_1 page of the PBW spectral sequence for $L_k(\mathfrak{g})$, the Poisson differential kills the root sequence for $L_k(\mathfrak{g})$, the Poisson differential pairs the root generators through $[e_\alpha, f_\alpha] = h_\alpha$, but the Cartan generators $h_1, \dots, h_r$ survive because $\mathfrak{h}$ is abelian. When $r = 1$, the surviving class is confined to a single column and cannot support a higher differential. When $r \geq 2$, the surviving r -dimensional space admits nontrivial higher differentials sourced by the multi-weight coupling between null vectors at different roots: the null vectors interact through the Cartan matrix, and the interactions generate off-diagonal bar cohomology.

Remark 7.4 (Evidence). For $\mathfrak{g} = \mathfrak{sl}_3$ at admissible level with $q = 3$: $\dim H^2(\bar{B}^{\text{ch}}(L_k(\mathfrak{sl}_3))) = 2$, confirmed by direct computation. The two classes correspond to the two simple roots. For $\mathfrak{g} = \mathfrak{sl}_4$: predicted $H^2 = 3$. The pattern is $\dim H^2 = \text{rank}(\mathfrak{g})$.

Remark 7.5 (Physical content). Admissible-level simple quotients are the vertex algebras of rational conformal field theories (Arakawa). If Conjecture 7.3 holds, then bar–cobar inversion fails for these quotients at rank ≥ 2 : the null vectors that enforce rationality of the representation category simultaneously obstruct PBW concentration. Koszulness and rationality would stand in tension for rank ≥ 2 : the algebraic optimality of the bar complex (Koszulness) is incompatible with the representation-theoretic rigidity of the simple quotient (rationality). At rank one, both coexist.

8. THE SHADOW ALGEBRA AND BV/BRST

The Koszul property controls the bar complex at genus zero. At higher genus, the bar differential squares to a nonzero curvature $d_{\text{fib}}^2 = \kappa(\mathcal{A}) \cdot \omega_g$, and the Maurer–Cartan element $\Theta_{\mathcal{A}}$ is the datum that measures the obstruction to extending genus-zero results across the modular tower. This section records two structures that accompany the Koszul property: the shadow algebra (the graded Lie algebra controlling the MC recursion) and the BV/bar comparison (the genus-by-genus identification with the BRST complex).

8.1. The shadow algebra. The shadow invariants $S_r(\mathcal{A})$ for $r \geq 2$ are the degree- r projections of $\Theta_{\mathcal{A}}$. They do not form a ring: the natural algebraic structure is a *graded Lie bracket*, inherited from the convolution Lie algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$.

Definition 8.1 (Shadow algebra). The *shadow algebra* of a chiral algebra $\mathcal{A}$ is the bigraded Lie algebra

$$\mathcal{A}^{\text{sh}} = \bigoplus_{r \geq 2, g \geq 0} \mathcal{A}_{r,g}^{\text{sh}}$$

where $\mathcal{A}_{r,g}^{\text{sh}}$ is the degree- r , genus- g component of the modular convolution algebra $\mathfrak{g}_{\mathcal{A}}^{\text{mod}}$. The Lie bracket $[-, -]^{\text{sh}}: \mathcal{A}_{r_1, g_1}^{\text{sh}} \otimes \mathcal{A}_{r_2, g_2}^{\text{sh}} \rightarrow \mathcal{A}_{r_1+r_2-2, g_1+g_2}^{\text{sh}}$ descends from the convolution bracket and drives the Maurer–Cartan recursion: S_{r+1} is determined by $S_2, \dots, S_r$ through

$$d S_{r+1} + \frac{1}{2} \sum_{\substack{r_1+r_2=r+2 \\ r_1, r_2 \geq 2}} [S_{r_1}, S_{r_2}]^{\text{sh}} = 0.$$

Remark 8.2 (Lie-algebra structure). The shadow algebra is a graded Lie algebra, *not* a graded-commutative ring. There is no shadow product $S_r \cdot S_s$; there is a shadow bracket $[S_r, S_s]^{\text{sh}}$. The distinction is structural: the Jacobi identity of $[-, -]^{\text{sh}}$ controls the solvability of the MC recursion, and the depth classification G/L/C/M records where this recursion first produces nonzero data.

The G/L/C/M classification follows from the shadow algebra:

- Class G ($r_{\max} = 2$): $S_r = 0$ for $r \geq 3$. The shadow algebra is abelian. Heisenberg, lattice algebras.
- Class L ($r_{\max} = 3$): $S_3 \neq 0$, $S_r = 0$ for $r \geq 4$. The cubic shadow detects the Lie bracket. Affine Kac–Moody at non-critical level.
- Class C ($r_{\max} = 4$): $S_3 \neq 0$, $S_4 \neq 0$, $S_r = 0$ for $r \geq 5$. Contact-term termination. The $\beta\gamma$ system.
- Class M ($r_{\max} = \infty$): $S_r \neq 0$ for infinitely many r . Virasoro, $\mathcal{W}$ -algebras.

Every class is chirally Koszul. The depth classification is orthogonal to Koszulness: it measures complexity within the Koszul world, not failure of Koszulness.

8.2. BV/bar identification in the coderived category. The BV/BRST complex $C_{\text{BV}}^{\bullet}(\mathcal{A}, \Sigma_g)$ on a genus- g surface Σ_g computes the same factorization homology as the bar complex $B^{(g)}(\mathcal{A})$, but through a different chain model: the BV propagator is the Green’s function on Σ_g , while the bar propagator is the logarithmic form $d \log E(z, w)$ (weight 1, the prime form logarithmic derivative).

Theorem 8.3 (BV=bar in the coderived category). *Let $\mathcal{A}$ be a chirally Koszul algebra.*

- For every genus $g \geq 0$, $C_{\text{BV}}^{\bullet}(\mathcal{A}, \Sigma_g) \simeq_{D^{\infty}} B^{(g)}(\mathcal{A})$ in the coderived category of curved factorization models.
- If $\mathcal{A}$ is class G or class L, the comparison morphism f_g is a chain-level weak equivalence.
- If $\mathcal{A}$ is class C and harmonic decoupling holds, f_g is a chain-level weak equivalence.
- If $\mathcal{A}$ is class M, the comparison cone is coacyclic: f_g is a coderived equivalence but not a chain-level quasi-isomorphism. The harmonic discrepancy at degree r is $\delta_r^{\text{harm}} = c_r(\mathcal{A}) \cdot m_0^{\lfloor r/2 \rfloor - 1}$, a sum of curvature powers whose cone is exact in the coderived sense.

Proof sketch. The comparison morphism f_g is constructed by replacing the BV propagator with its Hodge decomposition relative to the bar propagator. At genus 0, the Hodge decomposition is trivial (no harmonic forms on $\mathbb{P}^1$), and f_0 is a chain-level quasi-isomorphism for all classes.

At genus $g \geq 1$, the Hodge decomposition produces a harmonic correction proportional to the curvature $m_0 = \kappa(\mathcal{A}) \cdot \omega_g$. For class G, $\kappa = k$ and $S_r = 0$ for $r \geq 3$: the harmonic correction at each degree involves only the curvature itself, and the comparison is exact. For class L, the cubic shadow contributes a single additional correction at degree 3, which vanishes by the Jacobi identity of the shadow bracket.

For class M, the harmonic corrections are nonzero at every degree $r \geq 4$. Each correction δ_r^{harm} is a polynomial in the curvature m_0 ; the sum defines the comparison cone. This cone is *not* acyclic in the ordinary sense (the harmonic terms persist at each degree), but it *is* coacyclic: it belongs to the acyclic class of the coderived category. The proof uses the characterization of coacyclic objects as those whose totalizations are contractible in the complete filtered sense [24, §3.7]. $\square$

Remark 8.4 (Class M: chain-level false). The failure of the chain-level BV/bar comparison for class M is genuine: it is not an artifact of a suboptimal chain model. The Virasoro algebra has $\partial_4^{\text{harm}} = c_4(\text{Vir}_c) \cdot m_o \neq 0$ for $c \neq 0$, and c_4 is determined by the shadow invariant $S_4 = 10/[c(5c + 22)]$. Since $S_4 \neq 0$ for generic c , the harmonic discrepancy is nonzero, and no chain-level comparison exists. The coderived category is the correct ambient category for BV/bar at class M.

Remark 8.5 (Scope summary). $\text{BV}=\text{bar}$ in the coderived category is proved unconditionally for all chirally Koszul algebras. The chain-level strengthening holds for class G and class L (unconditionally), for class C (conditionally on harmonic decoupling), and fails for class M. The coderived formulation is the natural one: it accommodates the curvature $d_{\text{fib}}^2 = \kappa \cdot \omega_g$ inside the coderived acyclic class.

9. STRUCTURAL SUMMARY

The classical KZ criteria detect chiral Koszulness on the core PBW component. The supplementary criteria are chart functors attached to other associator base points. Their overlap is controlled by Tamarkin transfer on finite PBW-Artin components carrying the same Koszul witness.

For Virasoro at generic central charge, the Hodge–Betti chart U_{12} proves Koszulness using the Feigin–Fuchs BGG resolution and the Kac determinant. Bar–cobar inversion is not an input to that proof.

For Yangians, the associative Polishchuk–Positselski PBW theorem supplies an associative Koszul chart. On the comparison component there is an open embedding $\mathcal{M}_{\text{Kosz}}^{\text{assoc}}(Y) \hookrightarrow \mathcal{M}_{\text{Kosz}}^{\text{chiral}}(Y)$ (Theorem 2.11). Chart-surjectivity is an isomorphism at $\mathfrak{sl}_2$ and remains open in higher rank.

The Koszul locus is the locus on which the bar construction is invertible, the spectral sequence collapses, the resolution is linear, and the scalar projection of the degree-two Maurer–Cartan element is controlled by κ . Fourteen chart functors detect this locus on their stated components. The standard generic rows lie on the Koszul locus. Shadow depth (the G/L/C/M classification) is an orthogonal invariant: it classifies complexity within the Koszul world, not proximity to the Koszul boundary. The admissible-level question (whether simple quotients at rank ≥ 2 remain Koszul) and the $\mathcal{D}$ -module purity converse (whether Koszulness implies Hodge-theoretic purity beyond affine Kac–Moody) are the two open frontiers where the boundary of the Koszul locus has not yet been mapped.

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